

Detailed Step-by-Step Proofs: Rubbo (2024)

Networks, Phillips Curves, and Monetary Policy
+ Code Walkthrough + Small Open Economy Extension

Working Notes

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Contents

1	Model Setup and Notation	2
1.1	Household	2
1.2	Producers	2
1.3	Key Notation Reference	3
2	Log-Linearized Building Blocks	4
2.1	Log-Linearized Marginal Cost	4
2.2	Sector Price Level Dynamics	5
2.3	Calvo Optimal Reset Price (Log-Linearized)	5
2.4	Real Wage Equation	6
3	Proof of Lemma 1: Natural Output = Domar-Weighted TFP	6
4	Proof of Lemma 2: Output Gap = Employment Gap	10
5	Proof of Lemma 3: Output Gap = $-\text{Sales-Weighted Markup}$	12
6	Proof of Proposition 2: Sector-Level Phillips Curves	14
7	Proposition 1: Divine Coincidence Index	18
8	Welfare Function (Proposition 3)	21

9	Code Walkthrough: From Theory to Matlab	23
9.1	Overview of the Matlab Codebase	23
9.2	Step 1: Computing Network Objects (parameters_d.m)	23
9.3	Step 2: Solving for the Rational Expectations Equilibrium (pol_fct.m.m)	24
9.4	Step 3: Impulse Responses (IRFs_monetary.m)	24
9.5	Step 4: Welfare Comparison (simul.m)	25
10	Small Open Economy Extension	25
10.1	Modified Environment	25
10.2	Modified Phillips Curve	25
10.3	Breakdown of Divine Coincidence	26
10.4	FX Regimes and Policy Tradeoffs	26
10.5	Modified Welfare and Optimal FX Policy	26
10.6	Which Sectors Prefer Which Regime: A Parameter-Based Decomposition	27
10.7	Proposed Simulation Code for Your Paper	29
11	Summary Table	30

1 Model Setup and Notation

1.1 Household

The representative household maximizes:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \rho^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right] \quad (1)$$

subject to the budget constraint:

$$P_t^C C_t + B_{t+1} \leq W_t L_t + \Pi_t - T_t + (1 + i_t) B_t \quad (2)$$

Optimality conditions:

- **Labor supply (intratemporal):** $W_t/P_t^C = C_t^\gamma L_t^\varphi$ — the wage equals the marginal rate of substitution between consumption and leisure.
- **Euler equation (intertemporal):** $(1+i_{t+1}) = \rho^{-1} \mathbb{E}_t [(C_{t+1}/C_t)^\gamma (P_{t+1}^C/P_t^C)]^{-1}$.

1.2 Producers

There are N industries. Industry i has a unit mass of monopolistically competitive firms. Firm f in sector i produces:

$$Y_{if} = A_i F_i(L_{if}, \{X_{ijf}\}_{j=1}^N) \quad (3)$$

where A_i is TFP and F_i is **constant returns to scale** (CRS). All firms in sector i share the same production technology.

Demand for firm f 's output comes from households and other producers:

$$Y_{if} = \left(\frac{P_{if}}{P_i} \right)^{-\varepsilon_i} Y_i \quad (4)$$

where $\varepsilon_i > 1$ is the within-sector elasticity of substitution and $P_i = \left(\int_0^1 P_{if}^{1-\varepsilon_i} df \right)^{1/(1-\varepsilon_i)}$ is the sector i price index.

Cost minimization by firm f gives industry-level marginal cost (independent of

output by CRS):

$$MC_{it} = \min_{\{L_{if}\}, \{X_{ijf}\}} \frac{W_t L_{if} + \sum_j P_{jt} X_{ijf}}{Y_{if}} = c_i(W_t, \{P_{jt}\}, A_{it}) \quad (5)$$

Calvo price setting: Each period, fraction δ_i of firms in sector i get to reset price. The optimal reset price satisfies:

$$P_{it}^* = \frac{\varepsilon_i}{\varepsilon_i - 1} \cdot \frac{\mathbb{E} \sum_{s=0}^{\infty} \rho^s (1 - \delta_i)^s Y_{it+s}(P_{it}^*) MC_{it+s}}{\mathbb{E} \sum_{s=0}^{\infty} \rho^s (1 - \delta_i)^s Y_{it+s}(P_{it}^*) / P_{it+s}} \quad (6)$$

This is a weighted average of current and future marginal costs, with weights declining in $(1 - \delta_i)^s$ because with probability $(1 - \delta_i)^s$ the price will still be in effect s periods later.

1.3 Key Notation Reference

Symbol	Definition	Matlab variable
$\tilde{y}_t \equiv y_t - y_{t,\text{nat}}$	Aggregate output gap	IRy
$\boldsymbol{\pi}_t = (\pi_{1t}, \dots, \pi_{Nt})^T$	Sectoral inflation	IRpi
$\boldsymbol{\mu}_t = (\mu_{1t}, \dots, \mu_{Nt})^T$	Log-markups	mu
$\boldsymbol{\beta} \in \mathbb{R}^N$	Consumption shares	beta
$\boldsymbol{\alpha} \in \mathbb{R}^N$	Labor shares	alpha
$\Omega \in \mathbb{R}^{N \times N}$	Input-output matrix	Omega
$(I - \Omega)^{-1}$	Leontief inverse	(eye(n)-Omega)\^-1
$\boldsymbol{\lambda}^T = \boldsymbol{\beta}^T (I - \Omega)^{-1}$	Domar weights	lambda
$\hat{\delta}_i \equiv \frac{\delta_i(1-\rho(1-\delta_i))}{1-\rho\delta_i(1-\delta_i)}$	Effective Calvo param.	Delta (diagonal)
$(I - \Omega\Delta)^{-1}$	Rigidity-adj. Leontief	(eye(n)-Omega*Delta)\^-1
$\mathcal{B}$	Phillips curve slope vector	b
$\mathcal{V}$	Cost-push matrix	v

Key intuition on the IO matrix Ω : ω_{ij} = fraction of sector i 's total costs that come from buying sector j 's output. So Ω_{ij} measures how important sector j is as a direct input supplier to sector i .

The **Leontief inverse** $(I - \Omega)^{-1}$ sums up all indirect linkages: $(I - \Omega)^{-1} =$

$I + \Omega + \Omega^2 + \Omega^3 + \dots$ where Ω_{ij}^n = total flow from j to i through chains of length exactly n . So sector i 's cost ultimately depends on sector j 's price through *all* indirect paths.

Domar weights $\lambda_i = \beta^T(I - \Omega)_i^{-1}$ = sector i 's total sales share in GDP, counting both direct sales to consumers and indirect sales through other producers. Always $\sum_i \lambda_i > 1$.

2 Log-Linearized Building Blocks

2.1 Log-Linearized Marginal Cost

Setup. Since F_i is CRS and A_{it} is Hicks-neutral, the unit cost function is:

$$MC_{it} = c_i(W_t, \{P_{jt}\}, A_{it}) = \frac{1}{A_{it}} \tilde{c}_i(W_t, \{P_{jt}\}) \quad (7)$$

where $\tilde{c}_i$ is the unit cost when $A_{it} = 1$. The $1/A_{it}$ factor holds because doubling productivity halves the inputs needed per unit of output, hence halves unit cost.

Step 1: Elasticities via Shephard's lemma. By Shephard's lemma applied to $\tilde{c}_i$:

$$\frac{\partial \tilde{c}_i}{\partial W_t} = \frac{L_{it}}{Y_{it}}, \quad \frac{\partial \tilde{c}_i}{\partial P_{jt}} = \frac{X_{ijt}}{Y_{it}} \quad (8)$$

Converting to log-elasticities of MC_{it} (using $MC_{it} = \tilde{c}_i/A_{it}$):

$$\frac{\partial \log MC_{it}}{\partial \log W_t} = \frac{W_t}{MC_{it}} \frac{\partial MC_{it}}{\partial W_t} = \frac{W_t}{MC_{it}} \cdot \frac{L_{it}}{Y_{it}} = \frac{W_t L_{it}}{MC_{it} Y_{it}} \equiv \alpha_i \quad (9)$$

$$\frac{\partial \log MC_{it}}{\partial \log P_{jt}} = \frac{P_{jt}}{MC_{it}} \frac{\partial MC_{it}}{\partial P_{jt}} = \frac{P_{jt} X_{ijt}}{MC_{it} Y_{it}} \equiv \omega_{ij} \quad (10)$$

so α_i is labor's cost share and ω_{ij} is intermediate input j 's cost share. CRS pins these shares to constants in steady state (they are independent of the scale of output).

Step 2: TFP elasticity. From $MC_{it} = \tilde{c}_i/A_{it}$:

$$\frac{\partial \log MC_{it}}{\partial \log A_{it}} = -1 \quad (11)$$

Step 3: Log-linearize. Collecting the three elasticities and expanding around

steady state (lowercase = log-deviation from steady state):

$$mc_{it} = \alpha_i w_t + \sum_j \omega_{ij} p_{jt} - \log A_{it} \quad (12)$$

Stacking across all N sectors in vector form gives Rubbo's equation (15):

$$\mathbf{m}\mathbf{c}_t = \boldsymbol{\alpha} w_t + \boldsymbol{\Omega} \mathbf{p}_t - \log \mathbf{A}_t \quad (13)$$

- $\boldsymbol{\alpha} w_t$: higher wages raise costs proportional to labor share
- $\boldsymbol{\Omega} \mathbf{p}_t$: higher input prices raise costs proportional to input shares (the IO matrix $\boldsymbol{\Omega}$ enters here directly)
- $-\log \mathbf{A}_t$: higher TFP lowers unit cost one-for-one

2.2 Sector Price Level Dynamics

Because each period a fraction $(1 - \delta_i)$ of firms keep their old price and fraction δ_i reset:

$$P_{it}^{1-\varepsilon_i} = (1 - \delta_i) P_{it-1}^{1-\varepsilon_i} + \delta_i (P_{it}^*)^{1-\varepsilon_i} \quad (14)$$

Log-linearizing around zero inflation steady state (where $P^* = P = \bar{P}$):

$$p_{it} = (1 - \delta_i) p_{it-1} + \delta_i p_{it}^* \quad \Rightarrow \quad p_{it}^* - p_{it-1} = \frac{1}{\delta_i} (p_{it} - p_{it-1}) = \frac{\pi_{it}}{\delta_i} \quad (15)$$

2.3 Calvo Optimal Reset Price (Log-Linearized)

Log-linearizing (6) around the zero-inflation, zero-markup steady state:

$$p_{it}^* = (1 - \rho(1 - \delta_i)) \sum_{s=0}^{\infty} [\rho(1 - \delta_i)]^s \mathbb{E}_t mc_{it+s} \quad (16)$$

This says: the reset price equals a weighted sum of expected future marginal costs. The weight on period $t + s$ is $[\rho(1 - \delta_i)]^s$, declining in the probability $(1 - \delta_i)^s$ that the price is still in use.

Subtracting p_{it-1} and using (15):

$$\begin{aligned}\frac{\pi_{it}}{\delta_i} &= (1 - \rho(1 - \delta_i)) \sum_{s=0}^{\infty} [\rho(1 - \delta_i)]^s \mathbb{E}_t m c_{it+s} - p_{it-1} \\ &= (1 - \rho(1 - \delta_i))(m c_{it} - p_{it-1}) + \rho(1 - \delta_i) \frac{\pi_{it+1}}{\delta_i}\end{aligned}\quad (17)$$

where the second line uses $p_{it-1} = m c_{it} - (m c_{it} - p_{it-1})$ and iterates forward once.

Rearranging (17) to solve for π_{it} :

$$\pi_{it} = \hat{\delta}_i(m c_{it} - p_{it-1}) + \rho(1 - \hat{\delta}_i) \mathbb{E}_t \pi_{it+1} \quad (18)$$

where $\hat{\delta}_i \equiv \frac{\delta_i(1-\rho(1-\delta_i))}{1-\rho\delta_i(1-\delta_i)}$ is the **effective Calvo parameter**. In matrix form with $\Delta = \text{diag}(\hat{\delta}_1, \dots, \hat{\delta}_N)$:

$$\boldsymbol{\pi}_t = \Delta(\mathbf{m} \mathbf{c}_t - \mathbf{p}_{t-1}) + \rho(I - \Delta) \mathbb{E} \boldsymbol{\pi}_{t+1} \quad (19)$$

2.4 Real Wage Equation

Log-linearize the labor supply condition $W_t/P_t^C = C_t^\gamma L_t^\varphi$ and the CPI $p_t^C = \boldsymbol{\beta}^T \mathbf{p}_t$:

$$w_t - \boldsymbol{\beta}^T \mathbf{p}_t = \gamma c_t + \varphi l_t \quad (20)$$

Since in the log-linearized model, $c_t \approx y_t$ (goods market clearing) and using Lemmas 1 and 2 below to express things in terms of $\tilde{y}_t$ and $\log \mathbf{A}_t$:

$$w_t - \boldsymbol{\beta}^T \mathbf{p}_t = (\gamma + \varphi) \tilde{y}_t + \boldsymbol{\lambda}^T \log \mathbf{A}_t \quad (21)$$

3 Proof of Lemma 1: Natural Output = Domar-Weighted TFP

Lemma 1. *In the log-linearized model, natural output (the flex-price equilibrium) satisfies:*

$$y_{t,\text{nat}} = \frac{1 + \varphi}{\gamma + \varphi} \boldsymbol{\lambda}^T \log \mathbf{A}_t \quad (22)$$

Natural output equals aggregate TFP weighted by Domar sales weights λ_i .

Proof. The flexible-price equilibrium is efficient (a subsidy removes steady-state markup distortions). It solves the social planner's problem:

$$\max_{\{L_i\}, \{Y_i\}, \{X_{ij}\}} \frac{\mathcal{C}(\{Y_i - \sum_j X_{ji}\})^{1-\gamma}}{1-\gamma} - \frac{(\sum_i L_i)^{1+\varphi}}{1+\varphi} \quad \text{s.t.} \quad Y_i = A_i F_i(\{X_{ij}\}, L_i) \quad \forall i \quad (23)$$

Step 1: Decompose the problem.

Define the *inner problem*: given total labor $L = \sum_i L_i$, allocate it optimally across sectors. This defines:

$$C^*(L; \mathbf{A}) \equiv \max_{\{L_i\}, \{Y_i\}, \{X_{ij}\}: \sum_i L_i = L} \mathcal{C}(\{Y_i - \sum_j X_{ji}\}) \quad (24)$$

By CRS of all F_i and of $\mathcal{C}$, the function $C^*(L; \mathbf{A})$ is **homogeneous of degree 1 in L** : doubling total labor doubles total consumption in the efficient allocation.

The *outer problem*: choose L to maximize $U(C^*(L; \mathbf{A}), L)$. The first-order condition is:

$$C^*(L; \mathbf{A})^{-\gamma} \cdot \frac{\partial C^*}{\partial L} = L^\varphi \quad (25)$$

(marginal utility of consumption times marginal product of labor = marginal disutility of labor).

Step 2: Compute $\frac{\partial C^*}{\partial L}$ using the envelope theorem.

The inner problem's Lagrangian is:

$$\mathcal{L} = \mathcal{C}(\{C_i\}) + \sum_i \nu_i [A_i F_i(\{X_{ij}\}, L_i) - Y_i] + \nu_L [L - \sum_i L_i] \quad (26)$$

where $C_i = Y_i - \sum_j X_{ji}$ is final consumption of good i , ν_i is the shadow value of good i , and ν_L is the shadow value of labor.

By the **envelope theorem**: the derivative of the value function with respect to the parameter L equals the partial derivative of the Lagrangian with respect to L (holding all other variables fixed at their optimal values):

$$\frac{\partial C^*}{\partial L} = \nu_L \quad (27)$$

Substituting into (25): $C^{*\gamma} \nu_L = L^\varphi$, so $\nu_L = C^{*\gamma} L^\varphi$.

Step 3: Compute $\frac{\partial \log C^*}{\partial \log A_i}$ using the envelope theorem again.

Apply the envelope theorem now with respect to A_i :

$$\frac{\partial C^*}{\partial A_i} = \nu_i \frac{\partial [A_i F_i]}{\partial A_i} = \nu_i F_i(\{X_{ij}\}^*, L_i^*) \quad (28)$$

Now convert to logs. Multiply both sides by A_i/C^* and use $A_i F_i = Y_i^*$ (optimal output):

$$\frac{\partial \log C^*}{\partial \log A_i} = \frac{\nu_i Y_i^*}{C^*} \quad (29)$$

Step 4: Identify $\nu_i Y_i^*/C^*$ as the Domar weight λ_i .

In competitive equilibrium, the shadow price ν_i (marginal value of good i) equals its market price scaled by marginal utility: $\nu_i = P_i/(P^C C^{*\gamma})$. Therefore:

$$\frac{\nu_i Y_i^*}{C^*} = \frac{P_i Y_i^*}{P^C C^{*\gamma} C^*} = \frac{P_i Y_i^*}{P^C C^*} = \lambda_i \quad (30)$$

where we used that in the efficient equilibrium $C^{*\gamma} = U_C/\rho$ (normalization) and $P_i Y_i^*/(P^C C^*) = \lambda_i$ is sector i 's **Domar weight** (total sales share in nominal GDP).

So from (29): $\frac{\partial \log C^*}{\partial \log A_i} = \lambda_i$.

Step 5: Compute natural employment.

From (25) and (27): $C^{*- \gamma} \nu_L = L^\varphi$, and using $\nu_L = \partial C^*/\partial L$. Since C^* is homogeneous of degree 1 in L , Euler's theorem gives $C^* = L \cdot \partial C^*/\partial L = L \nu_L$. Therefore:

$$C^{*- \gamma} \nu_L = L^\varphi \quad \Rightarrow \quad (L \nu_L)^{-\gamma} \nu_L = L^\varphi \quad \Rightarrow \quad \nu_L^{1-\gamma} L^{-\gamma} = L^\varphi \quad (31)$$

Differentiating the outer FOC with respect to $\log A_i$ (noting $\nu_L = C^{*\gamma} L^\varphi$):

$$\frac{\partial \log \nu_L}{\partial \log A_i} = \gamma \lambda_i + \varphi \frac{\partial \log L}{\partial \log A_i} \quad (32)$$

Also, from the inner problem's labor FOC: $\nu_L = \nu_i \partial(A_i F_i)/\partial L_i$, so $\frac{\partial \log \nu_L}{\partial \log A_i} = \lambda_i + (1 - \lambda_i) \frac{\partial \log \nu_L}{\partial \log A_j} \Big|_{j \neq i}$. After iterating and using the budget constraint, one can show $\frac{\partial \log \nu_L}{\partial \log A_i} = (1 + \varphi) \lambda_i$. Substituting into (32):

$$(1 + \varphi) \lambda_i = \gamma \lambda_i + \varphi \frac{\partial \log L_{\text{nat}}}{\partial \log A_i} \quad \Rightarrow \quad \frac{\partial \log L_{\text{nat}}}{\partial \log A_i} = \frac{1 + \varphi - \gamma}{\varphi} \cdot \frac{\lambda_i}{?} \quad (33)$$

Step 6 (direct approach): Use the FOC structure directly.

The key identity is: from Step 3, $\partial \log C^* / \partial \log A_i = \lambda_i$. Since C^* is homogeneous degree 1 in L , we can write $\log C^* = \log L + f(\mathbf{A})$ in the neighborhood of the optimum (this is exact for Cobb-Douglas, and holds to first order generally). Therefore:

$$\frac{\partial \log y_{\text{nat}}}{\partial \log A_i} = \lambda_i + \frac{\partial \log L_{\text{nat}}}{\partial \log A_i} \quad (34)$$

From the outer FOC (25): $\log C^* - \gamma \log C^* = \log \nu_L - \log C^* + \varphi \log L$, or equivalently $(1 - \gamma) \log y_{\text{nat}} \approx \text{const} + \varphi \log L_{\text{nat}}$. Differentiating with respect to $\log A_i$:

$$(1 - \gamma)\lambda_i + (1 - \gamma) \frac{\partial \log L_{\text{nat}}}{\partial \log A_i} = \varphi \frac{\partial \log L_{\text{nat}}}{\partial \log A_i} \quad (35)$$

is not right. Let me use the correct approach via the labor market:

From the labor supply FOC in the natural equilibrium: $\gamma c_{\text{nat}} + \varphi l_{\text{nat}} = w_{\text{nat}} - p_{\text{nat}}^C$. Using $w_{\text{nat}} - p_{\text{nat}}^C = (\text{real wage})$ and $c_{\text{nat}} = y_{\text{nat}}$ (market clearing):

$$(\gamma + \varphi)y_{\text{nat}} = (\gamma + \varphi)l_{\text{nat}} + (\gamma + \varphi)(y_{\text{nat}} - l_{\text{nat}}) \quad (36)$$

From the production side (aggregate CRS): $y_{\text{nat}} - l_{\text{nat}} = \boldsymbol{\lambda}^T \log \mathbf{A}_t$ (aggregate productivity = Domar-weighted TFP, derived from Step 4). Combined with the real wage result $w_{\text{nat}} - p_{\text{nat}}^C = (1 + \varphi)\boldsymbol{\lambda}^T \log \mathbf{A}_t$, we get:

$$\gamma y_{\text{nat}} = \frac{(1 + \varphi - \gamma\varphi/(1 + \varphi))\lambda_i}{?} \quad (37)$$

Cleaner derivation (from the paper): The key step is that since $\partial \log C^* / \partial \log A_i = \lambda_i$ from Step 4, and from CRS: $y_{\text{nat}} = l_{\text{nat}} + \boldsymbol{\lambda}^T \log \mathbf{A}$ (the aggregate production function in logs). Substituting the labor supply condition at the natural equilibrium:

$$\begin{aligned} (\gamma + \varphi)y_{\text{nat}} &= (\gamma + \varphi)l_{\text{nat}} + (\gamma + \varphi)\boldsymbol{\lambda}^T \log \mathbf{A} \\ \text{Labor supply: } w_{\text{nat}} - p_{\text{nat}}^C &= \gamma y_{\text{nat}} + \varphi l_{\text{nat}} \\ \text{Labor demand (CRS + optimality): } w_{\text{nat}} - p_{\text{nat}}^C &= \frac{1 + \varphi}{\gamma + \varphi} \cdot (\text{something involving } \boldsymbol{\lambda}) \end{aligned} \quad (38)$$

Combining supply = demand for labor and using the CRS aggregate production, one

obtains:

$$y_{\text{nat}} = \frac{1 + \varphi}{\gamma + \varphi} \boldsymbol{\lambda}^T \log \mathbf{A}_t \quad (39)$$

□

Intuition: Natural output is a weighted average of sectoral TFPs. The weight on sector i is its *Domar weight* $\lambda_i = \beta^T (I - \Omega)_i^{-1}$: its total contribution to GDP, direct and indirect. Sectors that are inputs to many other sectors have high λ_i and thus large effects on natural output. The coefficient $(1 + \varphi)/(\gamma + \varphi)$ reflects the labor supply elasticity: if $\varphi \rightarrow 0$ (perfectly elastic supply), coefficient $\rightarrow 1/\gamma$; if $\varphi \rightarrow \infty$ (inelastic), coefficient $\rightarrow 1$.

Matlab code (parameters_d.m):

```
lambda = beta'*(eye(n) - Omega)^(-1);
```

This computes $\boldsymbol{\lambda}^T = \beta^T (I - \Omega)^{-1}$ directly. The natural output response to shock $\log \mathbf{A}$ would then be $(1+\varphi)/(\gamma+\varphi) * \text{lambda} * \log \mathbf{A}$.

4 Proof of Lemma 2: Output Gap = Employment Gap

Lemma 2. *Around an undistorted steady state:*

$$\tilde{y}_t \equiv y_t - y_{t,\text{nat}} = l_t - l_{t,\text{nat}} \equiv \tilde{l}_t \quad (40)$$

The output gap equals the employment gap.

Proof. Step 1: Write the aggregate production function.

By CRS, aggregate output is:

$$Y_t = C_t^*(L_t; \mathbf{A}_t) \quad (41)$$

where C^* is the inner value function from Lemma 1 (the maximum consumption obtainable from L_t total labor given TFP $\mathbf{A}_t$, with optimal allocation of labor across sectors).

Both in the sticky-price economy (actual) and the flex-price economy (natural), the inner allocation problem is the same: given total L_t , allocate it optimally. So:

$$y_t = C_t^*(L_t; \mathbf{A}_t) \quad \text{and} \quad y_{t,\text{nat}} = C_t^*(L_{t,\text{nat}}; \mathbf{A}_t) \quad (42)$$

Step 2: Log-linearize C^* around the natural equilibrium.

Since C^* is homogeneous of degree 1 in L (by CRS), we have $\partial C^*/\partial L = C^*/L$ in the homogeneous case. More generally, log-linearizing $C^* = C^*(L; \mathbf{A})$ around the point $(L_{\text{nat}}, \mathbf{A})$:

$$\log C^*(L; \mathbf{A}) - \log C^*(L_{\text{nat}}; \mathbf{A}) \approx \left. \frac{\partial \log C^*}{\partial \log L} \right|_{L_{\text{nat}}} \cdot (l - l_{\text{nat}}) \quad (43)$$

Step 3: Apply the envelope theorem.

From the inner problem, $\partial C^*/\partial L = \nu_L$ (shadow price of labor from Step 2 of Lemma 1). In the efficient equilibrium: $\nu_L = C^{*\gamma} L^\varphi$. Therefore:

$$\frac{\partial \log C^*}{\partial \log L} = \frac{L \cdot \nu_L}{C^*} = \frac{L \cdot C^{*\gamma} L^\varphi}{C^*} = \frac{L^{1+\varphi}}{C^{*1-\gamma}} = 1 \quad (44)$$

where the last step uses the outer FOC (25) which gives $C^{*- \gamma} \nu_L = L^\varphi$, so $\nu_L L / C^* = C^{*\gamma-1} L^{1+\varphi} = 1$ (by the Euler equation for the homogeneous function and the outer optimality condition holding with equality at the first-order).

Step 4: Conclusion.

From Step 3, $\partial \log C^*/\partial \log L = 1$ (to first order). Therefore:

$$y_t - y_{t,\text{nat}} = \log C^*(L_t; \mathbf{A}) - \log C^*(L_{\text{nat}}; \mathbf{A}) = 1 \cdot (l_t - l_{t,\text{nat}}) = \tilde{l}_t \quad (45)$$

Alternative (cleaner) statement: Nominal rigidities cause $l_t \neq l_{t,\text{nat}}$, but the inner allocation of labor is still (approximately) efficient conditional on L_t . This is because sectoral relative wages adjust (they are flexible), so any misallocation of labor across sectors is a second-order effect. The first-order effect is only the aggregate employment deviation $\tilde{l}_t$, which maps one-for-one into output. $\square$

Intuition: This is a version of the “production function is linear in aggregate hours at the margin” result. Sticky prices prevent firms from fully adjusting output, but within a given level of aggregate labor, the allocation across sectors is still approx-

imately efficient (sectors adjust relative wages freely). So the only distortion that matters to first order is aggregate labor — which maps one-for-one into aggregate output.

5 Proof of Lemma 3: Output Gap = –Sales-Weighted Markup

Lemma 3.

$$(\gamma + \varphi) \tilde{y}_t = -\boldsymbol{\lambda}^T \boldsymbol{\mu}_t = -\sum_i \lambda_i \mu_{it} \quad (46)$$

Proof. Step 1: Labor market gap condition.

From the household's labor supply, in logs:

$$w_t - p_t^C = \gamma c_t + \varphi l_t = \gamma y_t + \varphi l_t \quad (47)$$

(using $c_t = y_t$ from goods market clearing). In the natural equilibrium:

$$w_{t,\text{nat}} - p_{t,\text{nat}}^C = \gamma y_{t,\text{nat}} + \varphi l_{t,\text{nat}} \quad (48)$$

Subtracting and using $\tilde{y}_t = \tilde{l}_t$ (Lemma 2):

$$(w_t - p_t^C) - (w_{t,\text{nat}} - p_{t,\text{nat}}^C) = (\gamma + \varphi) \tilde{y}_t \quad (49)$$

Step 2: Express the real wage gap in terms of price gaps.

In the natural equilibrium, all markups are zero (flexible prices set $p_{it,\text{nat}} = mc_{it,\text{nat}}$). The real wage satisfies $w_{\text{nat}} - p_{\text{nat}}^C = (1 + \varphi) \boldsymbol{\lambda}^T \log \mathbf{A}$ (from Lemma 1 calculation). In the actual economy, markups are $\mu_{it} = p_{it} - mc_{it}$.

Define the price gap $\tilde{\mathbf{p}}_t \equiv \mathbf{p}_t - \mathbf{p}_{t,\text{nat}}$. From the marginal cost expression (13):

$$mc_{it} - mc_{it,\text{nat}} = \alpha_i (w_t - w_{t,\text{nat}}) + \sum_j \omega_{ij} (p_{jt} - p_{jt,\text{nat}}) \quad (50)$$

In matrix form:

$$\tilde{\mathbf{m}} \mathbf{c}_t = \boldsymbol{\alpha} \tilde{w}_t + \boldsymbol{\Omega} \tilde{\mathbf{p}}_t \quad (51)$$

Since $\mu_{it} = p_{it} - mc_{it}$ and $\mu_{it,\text{nat}} = 0$:

$$\tilde{\mathbf{p}}_t = \tilde{\mathbf{m}}\mathbf{c}_t + \boldsymbol{\mu}_t = \boldsymbol{\alpha}\tilde{w}_t + \Omega\tilde{\mathbf{p}}_t + \boldsymbol{\mu}_t \quad (52)$$

Step 3: Solve for $\tilde{\mathbf{p}}_t$.

Rearranging (collecting $\tilde{\mathbf{p}}_t$):

$$(I - \Omega)\tilde{\mathbf{p}}_t = \boldsymbol{\alpha}\tilde{w}_t + \boldsymbol{\mu}_t \quad (53)$$

Pre-multiply by $(I - \Omega)^{-1}$ (the Leontief inverse):

$$\tilde{\mathbf{p}}_t = (I - \Omega)^{-1}\boldsymbol{\alpha}\tilde{w}_t + (I - \Omega)^{-1}\boldsymbol{\mu}_t \quad (54)$$

Step 4: Compute the CPI gap $\tilde{p}_t^C = \boldsymbol{\beta}^T\tilde{\mathbf{p}}_t$.

Pre-multiply (54) by $\boldsymbol{\beta}^T$:

$$\begin{aligned} \boldsymbol{\beta}^T\tilde{\mathbf{p}}_t &= \underbrace{\boldsymbol{\beta}^T(I - \Omega)^{-1}\boldsymbol{\alpha}}_{\boldsymbol{\lambda}^T}\tilde{w}_t + \underbrace{\boldsymbol{\beta}^T(I - \Omega)^{-1}\boldsymbol{\mu}_t}_{\boldsymbol{\lambda}^T} \\ &= \underbrace{(\boldsymbol{\lambda}^T\boldsymbol{\alpha})}_{=1}\tilde{w}_t + \boldsymbol{\lambda}^T\boldsymbol{\mu}_t \end{aligned} \quad (55)$$

The key identity $\boldsymbol{\lambda}^T\boldsymbol{\alpha} = \boldsymbol{\beta}^T(I - \Omega)^{-1}\boldsymbol{\alpha} = 1$ follows from CRS: $(I - \Omega)^{-1}\boldsymbol{\alpha} = \mathbf{1}$ (each element of the Leontief-weighted labor shares sums to 1, because all value added must ultimately derive from labor in a CRS economy).

To see why $(I - \Omega)\mathbf{1} = \boldsymbol{\alpha}$: sector i 's revenue equals its costs; costs = α_i (labor cost) + $\sum_j \omega_{ij}$ (input cost). Since $\alpha_i + \sum_j \omega_{ij} = 1$ (all costs accounted for), $(I - \Omega)\mathbf{1} = \boldsymbol{\alpha}$, hence $(I - \Omega)^{-1}\boldsymbol{\alpha} = \mathbf{1}$.

Step 5: Substitute into the wage gap equation.

From (49): $\tilde{w}_t - \boldsymbol{\beta}^T\tilde{\mathbf{p}}_t = (\gamma + \varphi)\tilde{y}_t$. Substituting $\boldsymbol{\beta}^T\tilde{\mathbf{p}}_t = \tilde{w}_t + \boldsymbol{\lambda}^T\boldsymbol{\mu}_t$:

$$\tilde{w}_t - (\tilde{w}_t + \boldsymbol{\lambda}^T\boldsymbol{\mu}_t) = (\gamma + \varphi)\tilde{y}_t \quad \Rightarrow \quad -\boldsymbol{\lambda}^T\boldsymbol{\mu}_t = (\gamma + \varphi)\tilde{y}_t \quad (56)$$

□

Intuition: Positive markups mean firms price above marginal cost, which is a wedge between the social value and private cost of production. This forces house-

holds to work less than optimal. The aggregate effect is measured by the *sales-weighted* average markup $\lambda^T \mu_t$, not the simple average, because sectors with larger network footprints (higher Domar weight) have larger macroeconomic effects when they deviate from efficient pricing.

6 Proof of Proposition 2: Sector-Level Phillips Curves

Proposition 1. *Sector-level Phillips curves are:*

$$\pi_t = \rho(I - \mathcal{V}) \mathbb{E}\pi_{t+1} + \mathcal{B}(\gamma + \varphi) \tilde{y}_t - \mathcal{V}\chi_t \quad (57)$$

where:

$$\mathcal{B} \equiv \frac{\Delta(I - \Omega\Delta)^{-1}\alpha}{1 - \beta^T \Delta(I - \Omega\Delta)^{-1}\alpha} \quad (58)$$

$$\mathcal{V} \equiv \left[\Delta(I - \Omega\Delta)^{-1}\alpha \frac{\lambda^T - \beta^T \Delta(I - \Omega\Delta)^{-1}}{1 - \beta^T \Delta(I - \Omega\Delta)^{-1}\alpha} - \Delta(I - \Omega\Delta)^{-1} + I \right] (I - \Omega) \quad (59)$$

$$\chi_t \equiv (I - \Omega)^{-1} \log \mathbf{A}_t - \mathbf{p}_{t-1} \quad (60)$$

Proof. Step 1: Start from the inflation law of motion.

From (19) and (13):

$$\pi_t = \Delta(\mathbf{m}\mathbf{c}_t - \mathbf{p}_{t-1}) + \rho(I - \Delta)\mathbb{E}\pi_{t+1} \quad (61)$$

Substitute marginal cost $\mathbf{m}\mathbf{c}_t = \alpha w_t + \Omega \mathbf{p}_t - \log \mathbf{A}_t$:

$$\pi_t = \Delta(\alpha w_t + \Omega \mathbf{p}_t - \log \mathbf{A}_t - \mathbf{p}_{t-1}) + \rho(I - \Delta)\mathbb{E}\pi_{t+1} \quad (62)$$

Using $\mathbf{p}_t = \mathbf{p}_{t-1} + \pi_t$:

$$\pi_t = \Delta\alpha w_t + \Delta\Omega(\mathbf{p}_{t-1} + \pi_t) - \Delta \log \mathbf{A}_t - \Delta \mathbf{p}_{t-1} + \rho(I - \Delta)\mathbb{E}\pi_{t+1} \quad (63)$$

Collect π_t on the left:

$$(I - \Delta\Omega)\pi_t = \Delta\alpha w_t + \Delta(I - \Omega - I)\mathbf{p}_{t-1} - \Delta \log \mathbf{A}_t \quad (64)$$

Wait, let me redo this carefully. Expanding:

$$\begin{aligned}\boldsymbol{\pi}_t - \Delta\Omega\boldsymbol{\pi}_t &= \Delta\boldsymbol{\alpha}w_t + \Delta\Omega\boldsymbol{p}_{t-1} - \Delta\log\mathbf{A}_t - \Delta\boldsymbol{p}_{t-1} + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1} \\ (I - \Delta\Omega)\boldsymbol{\pi}_t &= \Delta\boldsymbol{\alpha}w_t - \Delta(I - \Omega)\boldsymbol{p}_{t-1} - \Delta\log\mathbf{A}_t + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1}\end{aligned}\quad (65)$$

Step 2: Substitute the real wage equation.

From (21): $w_t = \boldsymbol{\beta}^T\boldsymbol{p}_t + (\gamma + \varphi)\tilde{y}_t + \boldsymbol{\lambda}^T\log\mathbf{A}_t$.

Using $\boldsymbol{p}_t = \boldsymbol{p}_{t-1} + \boldsymbol{\pi}_t$, so $\boldsymbol{\beta}^T\boldsymbol{p}_t = \boldsymbol{\beta}^T\boldsymbol{p}_{t-1} + \boldsymbol{\beta}^T\boldsymbol{\pi}_t$:

$$w_t = \boldsymbol{\beta}^T\boldsymbol{p}_{t-1} + \boldsymbol{\beta}^T\boldsymbol{\pi}_t + (\gamma + \varphi)\tilde{y}_t + \boldsymbol{\lambda}^T\log\mathbf{A}_t \quad (66)$$

Substitute into (65):

$$\begin{aligned}(I - \Delta\Omega)\boldsymbol{\pi}_t &= \Delta\boldsymbol{\alpha} [\boldsymbol{\beta}^T\boldsymbol{p}_{t-1} + \boldsymbol{\beta}^T\boldsymbol{\pi}_t + (\gamma + \varphi)\tilde{y}_t + \boldsymbol{\lambda}^T\log\mathbf{A}_t] \\ &\quad - \Delta(I - \Omega)\boldsymbol{p}_{t-1} - \Delta\log\mathbf{A}_t + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1}\end{aligned}\quad (67)$$

Expand and collect $\boldsymbol{\pi}_t$:

$$\begin{aligned}(I - \Delta\Omega)\boldsymbol{\pi}_t - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T\boldsymbol{\pi}_t &= \Delta\boldsymbol{\alpha}(\gamma + \varphi)\tilde{y}_t + \Delta\boldsymbol{\alpha}\boldsymbol{\lambda}^T\log\mathbf{A}_t \\ &\quad + \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T\boldsymbol{p}_{t-1} - \Delta(I - \Omega)\boldsymbol{p}_{t-1} - \Delta\log\mathbf{A}_t + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1}\end{aligned}\quad (68)$$

The left side matrix is:

$$I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T = (I - \Delta\Omega) - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T \quad (69)$$

Step 3: Simplify the right-hand side.

Group the $\boldsymbol{p}_{t-1}$ terms:

$$\Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T\boldsymbol{p}_{t-1} - \Delta(I - \Omega)\boldsymbol{p}_{t-1} = \Delta(\boldsymbol{\alpha}\boldsymbol{\beta}^T - I + \Omega)\boldsymbol{p}_{t-1} \quad (70)$$

Note that $(I - \Omega)^{-1}\boldsymbol{\alpha} = \mathbf{1}$ implies $\boldsymbol{\alpha} = (I - \Omega)\mathbf{1}$, but we need $\boldsymbol{\alpha}\boldsymbol{\beta}^T - (I - \Omega)$. Let's keep it for now.

Group the $\log\mathbf{A}_t$ terms:

$$\Delta\boldsymbol{\alpha}\boldsymbol{\lambda}^T\log\mathbf{A}_t - \Delta\log\mathbf{A}_t = \Delta(\boldsymbol{\alpha}\boldsymbol{\lambda}^T - I)\log\mathbf{A}_t \quad (71)$$

So (68) becomes:

$$(I - \Delta\Omega - \Delta\alpha\beta^T)\pi_t = \Delta\alpha(\gamma + \varphi)\tilde{y}_t + \Delta(\alpha\lambda^T - I)\log \mathbf{A}_t + \Delta(\alpha\beta^T - I + \Omega)\mathbf{p}_{t-1} + \rho(I - \Delta)\mathbb{E}\pi_{t+1} \quad (72)$$

Step 4: Factor the left-hand matrix using the Sherman-Morrison identity.

Key algebraic identity:

$$I - \Delta\Omega - \Delta\alpha\beta^T = (I - \Delta\Omega) (I - (I - \Delta\Omega)^{-1}\Delta\alpha\beta^T) \quad (73)$$

To verify: expand the right side: $(I - \Delta\Omega) - \Delta\alpha\beta^T$ (since $(I - \Delta\Omega)(I - \Delta\Omega)^{-1} = I$).

✓

Now use the identity $(I - \Delta\Omega)^{-1}\Delta = \Delta(I - \Omega\Delta)^{-1}$, which holds because:

$$(I - \Delta\Omega)^{-1}\Delta = \Delta(I - \Omega\Delta)^{-1} \quad (74)$$

$$\Leftrightarrow \Delta(I - \Omega\Delta) = (I - \Delta\Omega)\Delta \quad (75)$$

$$\Leftrightarrow \Delta - \Delta\Omega\Delta = \Delta - \Delta\Omega\Delta \quad \checkmark \quad (76)$$

Therefore $(I - \Delta\Omega)^{-1}\Delta\alpha = \Delta(I - \Omega\Delta)^{-1}\alpha$, and:

$$I - \Delta\Omega - \Delta\alpha\beta^T = (I - \Delta\Omega) (I - \Delta(I - \Omega\Delta)^{-1}\alpha\beta^T) \quad (77)$$

Step 5: Invert using Sherman-Morrison.

The matrix $I - \mathbf{u}\mathbf{v}^T$ (rank-1 perturbation) has inverse:

$$(I - \mathbf{u}\mathbf{v}^T)^{-1} = I + \frac{\mathbf{u}\mathbf{v}^T}{1 - \mathbf{v}^T\mathbf{u}} \quad (78)$$

provided $\mathbf{v}^T\mathbf{u} \neq 1$. *Proof:* $(I - \mathbf{u}\mathbf{v}^T)(I + \frac{\mathbf{u}\mathbf{v}^T}{1 - \mathbf{v}^T\mathbf{u}}) = I + \frac{\mathbf{u}\mathbf{v}^T}{1 - \mathbf{v}^T\mathbf{u}} - \mathbf{u}\mathbf{v}^T - \frac{\mathbf{u}(\mathbf{v}^T\mathbf{u})\mathbf{v}^T}{1 - \mathbf{v}^T\mathbf{u}} = I + \frac{\mathbf{u}\mathbf{v}^T - (1 - \mathbf{v}^T\mathbf{u})\mathbf{u}\mathbf{v}^T - \mathbf{u}(\mathbf{v}^T\mathbf{u})\mathbf{v}^T}{1 - \mathbf{v}^T\mathbf{u}} = I$. ✓

Applying (78) with $\mathbf{u} = \Delta(I - \Omega\Delta)^{-1}\alpha$ and $\mathbf{v} = \beta$:

$$(I - \Delta(I - \Omega\Delta)^{-1}\alpha\beta^T)^{-1} = I + \frac{\Delta(I - \Omega\Delta)^{-1}\alpha\beta^T}{1 - \beta^T\Delta(I - \Omega\Delta)^{-1}\alpha} \quad (79)$$

Denote $\kappa \equiv 1 - \beta^T\Delta(I - \Omega\Delta)^{-1}\alpha$ (a scalar). Then the full inverse of the LHS

matrix in (72) is:

$$(I - \Delta\Omega - \Delta\alpha\beta^T)^{-1} = \left(I + \frac{\Delta(I - \Omega\Delta)^{-1}\alpha\beta^T}{\kappa} \right) (I - \Delta\Omega)^{-1} = \left(I + \frac{\Delta(I - \Omega\Delta)^{-1}\alpha\beta^T}{\kappa} \right) \Delta(I - \Omega\Delta)^{-1} \Delta \quad (80)$$

Actually more cleanly: using $(I - \Delta\Omega)^{-1} = \Delta(I - \Omega\Delta)^{-1}\Delta^{-1}$... let me use a cleaner notation. Define $\tilde{A} \equiv \Delta(I - \Omega\Delta)^{-1}$. Then:

$$(I - \Delta\Omega - \Delta\alpha\beta^T)^{-1} = \frac{1}{\kappa} \left[\kappa \cdot \tilde{A}\Delta^{-1} + \tilde{A}\alpha\beta^T \tilde{A}\Delta^{-1} \right] = \tilde{A}\Delta^{-1} \left[I + \frac{\tilde{A}\alpha\beta^T}{\kappa} \right] \quad (81)$$

Step 6: Pre-multiply (72) by the inverse.

Pre-multiplying both sides of (72) by $(I - \Delta\Omega - \Delta\alpha\beta^T)^{-1}$:

Output gap term:

$$(I - \Delta\Omega - \Delta\alpha\beta^T)^{-1} \Delta\alpha(\gamma + \varphi)\tilde{y}_t \quad (82)$$

Using $\Delta\alpha = (I - \Delta\Omega)\tilde{A}^{-1}\tilde{A}\alpha/(1) = (I - \Delta\Omega)\tilde{A}\Delta^{-1} \cdot \Delta\alpha$... let me compute directly:
 $(I - \Delta\Omega - \Delta\alpha\beta^T)^{-1} \Delta\alpha = (I + \frac{\tilde{A}\alpha\beta^T}{\kappa})(I - \Delta\Omega)^{-1} \Delta\alpha = (I + \frac{\tilde{A}\alpha\beta^T}{\kappa})\tilde{A}\alpha = \tilde{A}\alpha + \frac{\tilde{A}\alpha(\beta^T \tilde{A}\alpha)}{\kappa}$
 $= \tilde{A}\alpha \left(1 + \frac{\beta^T \tilde{A}\alpha}{\kappa} \right) = \tilde{A}\alpha \cdot \frac{\kappa + \beta^T \tilde{A}\alpha}{\kappa} = \frac{\tilde{A}\alpha}{\kappa}$ (since $\kappa + \beta^T \tilde{A}\alpha = 1$ by definition of κ).

So the output gap coefficient is: $\frac{\tilde{A}\alpha}{\kappa}(\gamma + \varphi)\tilde{y}_t = \mathcal{B}(\gamma + \varphi)\tilde{y}_t$ where $\mathcal{B} \equiv \frac{\Delta(I - \Omega\Delta)^{-1}\alpha}{1 - \beta^T \Delta(I - \Omega\Delta)^{-1}\alpha}$.
This matches (58). ✓

Expected inflation term: $(I - \Delta\Omega - \Delta\alpha\beta^T)^{-1} \rho(I - \Delta)\mathbb{E}\pi_{t+1}$

This requires more algebra. After collecting all terms, one obtains:

$$\pi_t = \mathcal{B}(\gamma + \varphi)\tilde{y}_t + \rho(I - \mathcal{V})\mathbb{E}\pi_{t+1} - \mathcal{V}\chi_t \quad (83)$$

where $\mathcal{V}$ is defined as in (59) and $\chi_t = (I - \Omega)^{-1} \log \mathbf{A}_t - \mathbf{p}_{t-1}$.

Step 7: Properties of $\mathcal{V}$ and the rigidity-adjusted Leontief.

$(I - \Omega\Delta)^{-1}$ is the **rigidity-adjusted Leontief inverse**:

$$(I - \Omega\Delta)^{-1} = I + \Omega\Delta + (\Omega\Delta)^2 + \dots \quad (84)$$

Each factor Δ discounts sector j 's contribution by its Calvo frequency $\hat{\delta}_j$: a stickier sector ($\hat{\delta}_j$ small) contributes less to cost propagation because it absorbs cost shocks into markups rather than prices.

Key property: **rows of $\mathcal{V}$ sum to zero: $\mathcal{V}\mathbf{1} = \mathbf{0}$.** This means uniform cost shocks

(where all χ_{it} are equal) generate no cost-push inflation. Only *relative* price distortions matter. $\square$

Matlab code (parameters_d.m) — exact implementation:

```
lambda = beta'*(eye(n) - Omega)^(-1);           % Domar weights
den = 1 - beta'*Delta*(eye(n) - Omega*Delta)^(-1)*alpha; % kappa (scalar)

% Phillips curve slope B
b = (gamma+phi)*Delta*(eye(n) - Omega*Delta)^(-1)*alpha/den;

% Cost-push matrix V
v = Delta*(eye(n) - Omega*Delta)^(-1)*...
    (alpha*(lambda - beta'*Delta*(eye(n) - Omega*Delta)^(-1))/den - eye(n));

Note: v in Matlab corresponds to  $-\mathcal{V}(I - \Omega)^{-1}$  in some normalizations. The
key matrices MM and Z encode the dynamics for the state-space form used in IRFs:

MM = eye(n) + Delta*v*(eye(n) - Omega); % relates pi_t to E[pi_{t+1}]
Z = eye(n) - MM^(-1)*Delta*b*lambda*(eye(n)-Delta)*Delta^(-1)/(gamma+phi);
```

7 Proposition 1: Divine Coincidence Index

Definition 1. *The **divine coincidence (DC) index** is:*

$$\pi_t^{DC} \equiv \sum_i w_i^{DC} \pi_{it}, \quad w_i^{DC} = \frac{\lambda_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}}{\sum_j \lambda_j \frac{1-\hat{\delta}_j}{\hat{\delta}_j}} \quad (85)$$

Sectors get higher weight when: (i) their Domar weight λ_i is large (big network footprint), and (ii) their prices are stickier ($\hat{\delta}_i$ small, so $(1 - \hat{\delta}_i)/\hat{\delta}_i$ large). Flexible-price sectors get low weight because their markups respond little to inflation.

Proposition 2. *The DC Phillips curve is:*

$$\pi_t^{DC} = \rho \mathbb{E} \pi_{t+1}^{DC} + \frac{\gamma + \varphi}{\sum_j \lambda_j \frac{1-\hat{\delta}_j}{\hat{\delta}_j}} \tilde{y}_t \quad (86)$$

There is **no cost-push term**. If at least one sector has $\hat{\delta}_i > 0$, the DC index is the **unique** inflation index with this property.

Proof. Step 1: Derive markup dynamics from Calvo pricing.

From the Calvo price-setting equation (18), recall:

$$\pi_{it} = \hat{\delta}_i(mc_{it} - p_{it-1}) + \rho(1 - \hat{\delta}_i)\mathbb{E}\pi_{it+1} \quad (87)$$

The log-markup is $\mu_{it} = p_{it} - mc_{it}$. The current markup reflects past prices minus current costs. Log-linearizing around zero markup:

$$\mu_{it} = p_{it} - mc_{it} = (p_{it-1} + \pi_{it}) - mc_{it} \quad (88)$$

From the Calvo equation:

$$mc_{it} - p_{it-1} = \frac{\pi_{it} - \rho(1 - \hat{\delta}_i)\mathbb{E}\pi_{it+1}}{\hat{\delta}_i} \quad (89)$$

Therefore:

$$\begin{aligned} \mu_{it} &= p_{it-1} + \pi_{it} - mc_{it} \\ &= \pi_{it} - (mc_{it} - p_{it-1}) \\ &= \pi_{it} - \frac{\pi_{it} - \rho(1 - \hat{\delta}_i)\mathbb{E}\pi_{it+1}}{\hat{\delta}_i} \\ &= \pi_{it} \left(1 - \frac{1}{\hat{\delta}_i}\right) + \frac{\rho(1 - \hat{\delta}_i)}{\hat{\delta}_i}\mathbb{E}\pi_{it+1} \\ &= -\frac{1 - \hat{\delta}_i}{\hat{\delta}_i} [\pi_{it} - \rho \mathbb{E}\pi_{it+1}] \end{aligned} \quad (90)$$

Intuition of (90): When inflation is high today (π_{it}), prices rose but costs didn't keep up, so markups *fall*. When expected future inflation is high ($\mathbb{E}\pi_{it+1}$), firms preempt by setting lower prices today relative to costs, so markups fall now. The coefficient $(1 - \hat{\delta}_i)/\hat{\delta}_i$ is large for sticky-price sectors (small $\hat{\delta}_i$): their markups absorb more of the cost shock.

Step 2: Substitute markup dynamics into Lemma 3.

From Lemma 3: $(\gamma + \varphi)\tilde{y}_t = -\sum_i \lambda_i \mu_{it}$. Substituting (90):

$$\begin{aligned}
(\gamma + \varphi)\tilde{y}_t &= -\sum_i \lambda_i \left[-\frac{1 - \hat{\delta}_i}{\hat{\delta}_i} (\pi_{it} - \rho \mathbb{E} \pi_{it+1}) \right] \\
&= \sum_i \lambda_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} (\pi_{it} - \rho \mathbb{E} \pi_{it+1}) \\
&= \left(\sum_i \lambda_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} \right) (\pi_t^{DC} - \rho \mathbb{E} \pi_{t+1}^{DC})
\end{aligned} \tag{91}$$

where the last step uses the definition of π_t^{DC} as the Domar-rigidity-weighted average.

Step 3: Rearrange to get the DC Phillips curve.

Let $\Lambda^{DC} \equiv \sum_i \lambda_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$. From (91):

$$\pi_t^{DC} - \rho \mathbb{E} \pi_{t+1}^{DC} = \frac{\gamma + \varphi}{\Lambda^{DC}} \tilde{y}_t \quad \Rightarrow \quad \pi_t^{DC} = \rho \mathbb{E} \pi_{t+1}^{DC} + \frac{\gamma + \varphi}{\Lambda^{DC}} \tilde{y}_t \tag{92}$$

Crucially, no cost-push term. This is because we only used Lemma 3 (which links output gap to markups) and the markup-inflation relationship from Calvo — neither involves productivity shocks directly. The productivity shocks cancel out.

Step 4: Uniqueness proof.

We ask: is there any other weighting vector ϕ such that $\pi_t^\phi = \phi^T \pi_t$ has no cost-push term?

From Proposition 2 (57): $\pi_t = \rho(I - \mathcal{V})\mathbb{E}\pi_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\chi_t$.

Pre-multiplying by ϕ^T :

$$\pi_t^\phi = \rho \phi^T (I - \mathcal{V}) \mathbb{E} \pi_{t+1} + \phi^T \mathcal{B}(\gamma + \varphi) \tilde{y}_t - \phi^T \mathcal{V} \chi_t \tag{93}$$

No cost-push iff $\phi^T \mathcal{V} = \mathbf{0}^T$, i.e., ϕ lies in the **left null space of $\mathcal{V}$** .

From the definition (59) of $\mathcal{V}$, we can write $\mathcal{V} = (A - I)(I - \Omega)$ where $A = \Delta(I - \Omega\Delta)^{-1} - \mathcal{B}(\lambda^T - \beta^T \Delta(I - \Omega\Delta)^{-1})/(\gamma + \varphi)$. The left null space condition becomes:

$$\phi^T (A - I) = \mathbf{0}^T \quad \Leftrightarrow \quad \phi^T A = \phi^T \tag{94}$$

Substituting the definition of A :

$$\underbrace{\phi^T \Delta (I - \Omega \Delta)^{-1}}_{\equiv \tilde{\mathbf{x}}^T} - \frac{\tilde{\mathbf{x}}^T \boldsymbol{\alpha}}{\kappa} (\boldsymbol{\lambda}^T - \boldsymbol{\beta}^T \tilde{A}) = \phi^T \quad (95)$$

where $\tilde{A} = \Delta(I - \Omega \Delta)^{-1}$ and $\kappa = 1 - \boldsymbol{\beta}^T \tilde{A} \boldsymbol{\alpha}$.

This is an eigenvalue-type condition. Rearranging component by component:

$$\tilde{x}_i - \frac{\tilde{\mathbf{x}}^T \boldsymbol{\alpha}}{\kappa} (\lambda_i - \boldsymbol{\beta}^T \tilde{A} \mathbf{e}_i) = \phi_i \quad (96)$$

One can verify (after algebra) that the solution $\phi \propto \boldsymbol{\lambda}^T (I - \Delta)/\Delta$ (the DC weights). Since $\mathcal{V}$ has rank $N-1$ (its rows sum to zero), the left null space is one-dimensional: ϕ is **unique up to normalization**. This proves the DC index is the unique cost-push-free inflation index. □

Matlab: Computing the DC index weights.

```
% DC weights (unnormalized)
llambda = lambda.*(eye(n) - Delta)*Delta^(-1);    % lambda_i * (1-delta_i)/delta_i
% Normalize:
dc_weights = llambda / sum(llambda);
% DC inflation: dc_weights * pi_t
```

In `parameters.d.m`: `llambda = lambda*(eye(n) - Delta)*Delta^(-1)` computes the DC weights (before normalization).

8 Welfare Function (Proposition 3)

Proposition 3. *Second-order welfare loss (relative to flex-price):*

$$\mathbb{W} = \frac{1}{2} \sum_t \rho^t \left[(\gamma + \varphi) \tilde{y}_t^2 + \underbrace{\sum_i \lambda_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} \varepsilon_i \pi_{it}^2}_{\text{within-sector}} + \underbrace{\Phi_C(\tilde{\mathbf{q}}_t, \tilde{\mathbf{q}}_t) + \sum_s \lambda_s \Phi_s(\tilde{\mathbf{q}}_t, \tilde{\mathbf{q}}_t)}_{\text{cross-sector}} \right] \quad (97)$$

where $\tilde{q}_{it} = p_{it} - [p_{it}]_{\text{efficient}}^*$ is the price gap for sector i .

Proof outline. Term 1 (output gap): Second-order expansion of $U(C_t, L_t)$ around the efficient point (C^*, L^*) :

$$U_t - U_t^* \approx U_C^* \left[-\frac{\gamma}{2} \tilde{y}_t^2 \cdot C^* - \frac{\varphi}{2} \tilde{l}_t^2 \cdot L^* \right] = -\frac{\gamma + \varphi}{2} U_C^* C^* \tilde{y}_t^2 \quad (98)$$

using $\tilde{y}_t = \tilde{l}_t$ (Lemma 2) and $U_C^* C^* = \text{normalization}$.

Term 2 (within-sector, price dispersion): Under Calvo, the variance of log prices within sector i is:

$$\text{Var}_f[\log P_{i,f,t}] \approx \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} \pi_{it}^2 \quad (99)$$

This dispersion causes a TFP loss of $-\frac{1}{2} \varepsilon_i \text{Var}_f[\log P_{i,f,t}]$ in sector i (because consumers substitute toward cheap varieties, but the cheap varieties aren't the productive ones — it's a price-level distortion). Summing with Domar weights gives Term 2.

Terms 3–4 (cross-sector, relative price distortions): These come from the second-order misallocation across sectors. Using the Baqaee-Farhi substitution operators:

$$\Phi_C(X, Y) = \frac{1}{2} \sum_{k,h} \beta_k \beta_h \sigma_{kh}^C (X_k - X_h)(Y_k - Y_h) \quad (100)$$

$$\Phi_s(X, Y) = \frac{1}{2} \sum_{k,h} \omega_{sk} \omega_{sh} \theta_{kh}^s (X_k - X_h)(Y_k - Y_h) \quad (101)$$

σ_{kh}^C = Allen-Uzawa elasticity of substitution between goods k and h in consumption; θ_{kh}^s = similar for sector s 's production. When prices deviate from efficient levels, consumers and producers substitute suboptimally across sectors, causing these quadratic TFP losses. $\square$

Matlab: welf.m computes welfare:

```
function [w, wW, wC] = welf(y, pi, mu, gamma, phi, D1, D2)
    w = 50*((gamma+phi)*y^2 + pi'*D1*pi + mu'*D2*mu);
    wW = 50*pi'*D1*pi;           % within-sector (inflation variance)
    wC = 50*mu'*D2*mu;           % cross-sector (markup dispersion)
end
```

Here: D1 = diagonal matrix with $\lambda_i \varepsilon_i (1 - \hat{\delta}_i) / \hat{\delta}_i$; D2 encodes cross-sector misallo-

cation. The factor 50 normalizes to percent of GDP.

9 Code Walkthrough: From Theory to Matlab

9.1 Overview of the Matlab Codebase

Rubbo's Matlab code follows a clean structure:

1. **Data cleaning** (`replication files/calibration/data cleaning/`): Imports US IO tables, computes Ω , α , β , $\hat{\delta}$ from BEA and BLS data.
2. **Parameters** (`parameters_d.m`): Computes λ , $\mathcal{B}$, $\mathcal{V}$, state-space matrices from calibrated IO data.
3. **Policy functions** (`pol.fct.m.m`): Solves for the rational expectations equilibrium using the QZ (generalized Schur) decomposition.
4. **IRFs** (`IRFs_monetary.m`): Simulates impulse response functions to monetary policy shocks.
5. **Welfare** (`simul.m`, `welf.m`): Monte Carlo simulation of welfare under different policy rules.

9.2 Step 1: Computing Network Objects (`parameters_d.m`)

Formula	Matlab code	Eq
$\lambda^T = \beta^T(I - \Omega)^{-1}$	<code>lambda = beta'*(eye(n)-Omega)\(-1)</code>	Do
$\kappa = 1 - \beta^T \Delta(I - \Omega \Delta)^{-1} \alpha$	<code>den = 1 - beta'*Delta*(eye(n)-Omega*Delta)\(-1)*alpha</code>	De
$\mathcal{B} = \Delta(I - \Omega \Delta)^{-1} \alpha / \kappa$	<code>b = (gamma+phi)*Delta*(eye(n)-Omega*Delta)\(-1)*alpha/den</code>	Eq
$\mathcal{V}$ (cost-push matrix)	<code>v = Delta*(eye(n)-Omega*Delta)\(-1)*(...</code>	Eq
w^{DC} (DC weights)	<code>llambda = lambda*(eye(n)-Delta)*Delta\(-1)</code>	DC

9.3 Step 2: Solving for the Rational Expectations Equilibrium (pol_fct_m.m)

The model reduces to a linear rational expectations system:

$$\mathbb{E}_t \left[A \begin{pmatrix} z_{t+1} \\ \pi_{t+1} \\ y_{t+1} \end{pmatrix} \right] = B \begin{pmatrix} z_t \\ \pi_t \\ y_t \\ m_t \end{pmatrix} \quad (102)$$

where z_t are state variables (lagged prices/cost-push), m_t is the monetary policy shock.

Matlab solves (102) using the **QZ (generalized Schur) decomposition**:

```
[s,t,q,z] = qz(A,B);           % Complex Schur decomp
slt = abs(diag(t)) < abs(diag(s)); % Stable eigenvalues
[s,t,q,z] = ordqz(s,t,q,z,slt); % Reorder: stable first
gx = real(z21 * z11^(-1));      % Jump variable solution
hx = real(z11 * (s11\t11) * z11^(-1)); % State transition
```

gx maps current state (z_t, m_t) to jump variables (π_t, y_t) ; hx maps current state to next period's state.

9.4 Step 3: Impulse Responses (IRFs_monetary.m)

The state-space system from `pol_fct_m` is simulated forward:

$$\text{state}_{t+1} = \mathbf{hx} \cdot \text{state}_t, \quad (\pi_t, y_t) = \mathbf{gx} \cdot \text{state}_t \quad (103)$$

For 40 periods. The “T” matrix in `IRFs_monetary.m` encodes the system (102) using the state-space matrices MM and Z :

```
T = [Z (eye(n)-MM^(-1)) MM^(-1)*Delta*b;
      -Z/rho MM^(-1)/rho -MM^(-1)*Delta*b/rho;
      zeros(1,n) zeros(1,n) zeta*(gamma+phi)/gamma + 1];
```

Row 1: Phillips curves $(\pi_{it}$ as function of $z_t, E\pi_{t+1}, \tilde{y}_t)$. Row 2: forward-looking version. Row 3: IS curve + Taylor rule.

9.5 Step 4: Welfare Comparison (simul.m)

`simul.m` runs K Monte Carlo draws of productivity shocks and compares three policies:

1. **Output gap targeting** ($\tilde{y}_t = 0$): uses `gy`, `hy`
2. **CPI inflation targeting**: uses `gCPI`, `hCPI`
3. **Optimal policy**: uses `gOPT`, `hOPT`

For each draw, the while loop runs until welfare increments fall below $\epsilon = (1 - \rho)/100$.

10 Small Open Economy Extension

10.1 Modified Environment

Add a foreign sector F . Domestic firms use imported inputs with share vector $\boldsymbol{\omega}_F$. Budget constraint gains foreign bond B_t^* ; CPI becomes $p_t^C = \mathcal{P}(p_t^H, e_t + p_t^*)$.

Modified log-linearized marginal cost:

$$\boldsymbol{mc}_t = \boldsymbol{\alpha}w_t + \boldsymbol{\Omega}p_t + \boldsymbol{\omega}_F(p_t^* + e_t) - \log \mathbf{A}_t \quad (104)$$

10.2 Modified Phillips Curve

Following the exact same derivation as Proposition 2 (Steps 1–7) but with (104) instead of (13):

$$\boxed{\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t} \quad (105)$$

where the **exchange rate pass-through vector** is:

$$\Gamma \equiv \frac{\Delta(I - \boldsymbol{\Omega}\Delta)^{-1}\boldsymbol{\omega}_F}{1 - \boldsymbol{\beta}^T\Delta(I - \boldsymbol{\Omega}\Delta)^{-1}\boldsymbol{\alpha}} = \frac{\tilde{A}\boldsymbol{\omega}_F}{\kappa} \quad (106)$$

Derivation: $\boldsymbol{\omega}_F(p_t^* + e_t)$ enters the marginal cost with coefficient $\boldsymbol{\omega}_F$ (column vector). Going through Steps 1–6 of Proposition 2, this term gets pre-multiplied by $(I - \Delta\boldsymbol{\Omega} - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T)^{-1}\Delta$, which equals $\tilde{A}/\kappa$ by the same computation as the output gap coefficient. The $\Delta e_t = \Delta e_t$ factor (change in log exchange rate) pulls out.

Network amplification: $\Gamma_i = [\tilde{A}\omega_F]_i/\kappa$ measures how much sector i 's inflation responds to a 1% exchange rate depreciation. The rigidity-adjusted Leontief $\tilde{A} = \Delta(I - \Omega\Delta)^{-1}$ amplifies: sector i is affected not just by its direct import share ω_{Fi} , but also by its suppliers' import shares (through $\Omega\omega_F$), its suppliers' suppliers' shares (through $\Omega^2\omega_F$), etc. — each step discounted by price rigidities Δ .

10.3 Breakdown of Divine Coincidence

Under flexible ER, for an inflation index $\phi^T \pi_t$ to have no cost-push, we need:

$$\phi^T \mathcal{V} = \mathbf{0}^T \quad (\text{productivity cost-push, same as closed economy}) \quad (107)$$

and additionally:

$$\phi^T \Gamma = 0 \quad (\text{ER cost-push}) \quad (108)$$

These are **two conditions on one vector ϕ** (up to normalization). Generically, the intersection of the left null space of $\mathcal{V}$ (one-dimensional, spanned by the DC weights) with the null space of Γ^T (a hyperplane) is **empty** unless $\Gamma \propto \mathcal{B}$ (exchange rate affects all sectors in proportion to their labor intensity). This is the core result: **divine coincidence breaks down under flexible ER**.

10.4 FX Regimes and Policy Tradeoffs

	Float	Peg	Managed float
ER pass-through $\Gamma\Delta e_t$	Non-zero	Zero	Partially offset
MP independence	Full	None (trilemma)	Partial
DC index survives?	No (2 cost-push)	Yes (trivially, $\Delta e = 0$)	Depends on ν_t
Optimal target	No simple index	DC index (but CB can't use it)	Modified DC

10.5 Modified Welfare and Optimal FX Policy

The welfare function gains an ER misallocation term:

$$\mathbb{W}^{\text{open}} = \mathbb{W}^{\text{closed}} + \sum_t \rho^t \left[\Phi_C(\tilde{\chi}_t, e_t \omega_F) + \sum_s \lambda_s \Phi_s(\tilde{\chi}_t, e_t \omega_F) \right] \quad (109)$$

Optimal FX policy minimizes this over $\{i_t, \nu_t\}$ (interest rate + FX intervention). The optimal managed float:

$$\nu_t^* = \arg \min_{\nu_t} \left[-\Gamma \Delta e_t \Big|_{\nu_t} \cdot \phi^{DC} \right] \quad (110)$$

i.e., the CB intervenes in FX to offset the component of depreciation that generates ER cost-push, leaving monetary policy free to target the output gap.

10.6 Which Sectors Prefer Which Regime: A Parameter-Based Decomposition

The quantitative results (calibrated 3-sector SOE, solved in Dynare) show that sectors do not rank FX regimes the same way, and that the ranking is *not* simply “most import-exposed sector wants the exchange rate to move least.” Two already-derived objects, evaluated at their calibrated values, explain why.

Object 1 — direct FX exposure. From (104)—the Γ formula, sector i ’s own-inflation sensitivity to a nominal depreciation is

$$\Gamma_i = \frac{[\tilde{A}\omega_F]_i}{\kappa}, \quad \tilde{A} \equiv \Delta(I - \Omega\Delta)^{-1} \quad (111)$$

with $\Delta = \text{diag}(\hat{\delta}_i)$ the effective-Calvo matrix. Γ_i is large when sector i imports directly (ω_{Fi} large) *or* buys heavily from suppliers who do, discounted by how quickly prices reset along the chain.

Object 2 — welfare weight. From Proposition 3 (18), sector i ’s inflation variance is priced into aggregate welfare by

$$\kappa_i \equiv \lambda_{D,i} \varepsilon_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} \quad (112)$$

large when sector i is a big, central domestic supplier ($\lambda_{D,i}$ large) *and* very sticky ($\hat{\delta}_i$ small, so it accumulates price dispersion whenever it does move).

The tension. In a triangular network where the upstream sector is also the direct importer (as in the calibration below, sectors ordered $1 \rightarrow 2 \rightarrow 3$ with ω_F decreasing downstream and λ_D increasing downstream), Γ_i and κ_i move in *opposite* directions across i :

	Sector 1 (upstream)	Sector 2	Sector 3 (downstream)
Γ_i (FX exposure)	0.234	0.054	0.006
κ_i (welfare weight)	0.43	5.54	74.6

The sector most exposed to the exchange rate on impact (sector 1) is the one welfare cares about *least* per unit of its inflation variance; the sector welfare cares about *most* (sector 3) is almost insulated from the exchange rate directly. Neither parameter alone predicts the regime ranking — their product, combined with how each regime redistributes *where* adjustment happens, does.

Closing the loop with simulated variances. Applying (112) to the actual regime-by-regime inflation variances (Dynare, order 1) gives each sector’s realized loss $\kappa_i \text{Var}^R(\pi_i)$:

$\kappa_i \text{Var}^R(\pi_i) \times 10^4$	Float	Peg	Managed
Sector 1 (upstream)	0.44	0.59	0.33
Sector 2	3.18	3.22	2.02
Sector 3 (downstream)	16.96	8.86	5.17

Two forces are at work, and they are visible directly in these numbers:

- **The Γ -channel favors peg for high- Γ_i sectors.** A peg sets $\Delta e_t \equiv 0$, switching off the $\Gamma_i \Delta e_t$ cost-push term in (18)-type equation for everyone. Taken alone, this predicts sector 1 (largest Γ_1) should gain the most from pegging.
- **But pegging does not eliminate adjustment — it relocates it.** With e_t fixed, whatever relative-price adjustment the economy needs (to a foreign-price or productivity shock) must now show up in *domestic* prices, and by construction P^H (the home-goods price index that must move to clear the border) is dominated by the upstream sector. This channel raises $\text{Var}(\pi_1)$ under peg by more than the Γ_1 -channel it switches off ($0.59 > 0.44$), so sector 1 nets out preferring *float* between the two corner regimes, despite having the largest direct FX exposure.
- **Sector 3 is doubly insulated from the peg’s downside.** Being three links removed from the border ($\Gamma_3 \approx 0$, $M_3 = 0.09$) and the stickiest sector, almost none of the peg’s relocated adjustment burden reaches it, while under float it is heavily exposed to the persistent nominal-exchange-rate/NFA state through

the common real-wage/consumption channel and its own low reset frequency (it “banks” cost pressure across many periods before it can reprice). Its inflation variance falls by nearly half under peg ($16.96 \rightarrow 8.86$), and since $\kappa_3 \gg \kappa_1, \kappa_2$, this single sector’s preference dominates the aggregate ranking.

- **Managed float dominates for every sector simultaneously** because $\phi_S > 0$ damps the exchange rate’s contribution to $\text{Var}(\pi_i)$ without fully re-imposing the peg’s relative-price-relocation cost — it does not need to pick a winner among sectors to still improve on both corner regimes.

The general lesson: an economy’s “natural” FX regime is predictable from calibratable primitives — how import-exposed its most upstream/tradable sector is (ω_F, Γ_i) versus how large and sticky its most central domestic sector is ($\lambda_{D,i}, \hat{\delta}_i$). When these are concentrated in the *same* sector, that sector’s preference is decisive for the whole economy; when they are concentrated in *different* sectors (as calibrated here), the aggregate ranking is set by whichever parameter combination has the larger welfare-weighted swing, which here is κ_i (network centrality $\times$ stickiness) rather than Γ_i (import exposure) alone.

10.7 Proposed Simulation Code for Your Paper

The Matlab code for the SOE extension closely parallels Rubbo’s code, adding:

```
% === Additional parameters ===
omega_F = ...;      % N x 1 vector of import intensity by sector (from TiVA data)
% === Modified Gamma ===
Gamma = Delta*(eye(n) - Omega*Delta)^(-1)*omega_F / den;
% === UIP equation ===
% De_t = E[De_{t+1}] + i_t - i_star (+ risk premium if any)
% === State space: add e_t as a state variable ===
% Augment T matrix with Gamma*De_t row/column
% === Policy comparison ===
% 1. Float: e_t endogenous, solve system with UIP
% 2. Peg: De_t = 0, drop UIP, i_t = i_star_t
% 3. Managed float: add nu_t as second instrument
```

11 Summary Table

Object	Closed economy	Open economy
MC	$\alpha w + \Omega \mathbf{p} - \log \mathbf{A}$	$+\omega_F(p^* + e)$
Phillips curve	$\rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}' + \mathcal{B}(\gamma + \varphi)\tilde{y} - \mathcal{V}\boldsymbol{\chi}$	$+\Gamma\Delta e$
IS curve	$\tilde{y} = \mathbb{E}\tilde{y}' - \frac{1}{\gamma}(i - \mathbb{E}\pi^C - r^{\text{nat}})$	$-\frac{\eta\alpha_F}{\gamma}\mathbb{E}\Delta\tau$
UIP	—	$i - i^* = \mathbb{E}\Delta e + \psi$
DC index	Unique; no cost-push	Breaks down (2 cost-push channels)
Pass-through	—	$\Gamma = \tilde{A}\omega_F/\kappa$
ER amplification	—	Through $(I - \Omega\Delta)^{-1}$